

Conjecture 00000004044 is false: HH^1 of the Kronecker quiver K_3 is not solvable

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1 The conjecture and its reading

Conjecture 00000004044 (English text):

Definition: HH^1 is the Lie algebra of outer derivations, the first Hochschild cohomology.

Conjecture: The HH^1 of a quiver algebra is solvable if and only if its oriented cycles are all broken after removing at most one vertex; the decision table is complete for $n \leq 6$ and the formula holds for all n .

The Chinese title also says “solvability criterion for HH^1 ” (*ke jie xing*), but the Chinese body uses the word *ke yue* (“reducible”) in the biconditional; see Remark 10.

Reading. For a quiver Q and a field k , the claim is

$\mathrm{HH}^1(kQ/I)$ is solvable $\iff$ some set of at most one vertex meets every oriented cycle of Q .

We refute the direction “ $\Leftarrow$ ” with a single quiver algebra, over every field k , with $n = 2$ vertices (so it lies inside the range $n \leq 6$ of the claimed decision table).

Counterexample. The generalized Kronecker quiver K_3 has vertices $1, 2$ and three parallel arrows $\alpha_0, \alpha_1, \alpha_2 : 1 \rightarrow 2$. It has no oriented cycle, so the cycle condition holds with zero vertices removed. Its path algebra kK_3 (ideal $I = 0$) is a quiver algebra in every usual sense: it is finite dimensional, and since K_3 has no path of length 2, $I = 0 = R_Q^2$ is an admissible ideal. We prove $\mathrm{HH}^1(kK_3)$ is not solvable. (It is known that $\mathrm{HH}^1(kK_3) \cong \mathfrak{pgl}_3(k)$; see the quote from [LRW] below. We do not use this isomorphism; we prove non-solvability directly.)

2 Definitions (as formalized)

All definitions are in the Lean file `Conjecture4044/Basic.lean` (namespace `C4044`).

Definition 1 (Derivations, inner derivations, HH^1). Let A be an associative unital algebra over a commutative ring k . $\mathbf{Der} \ k \ A$ is the set of k -linear maps $D : A \rightarrow A$ with $D(xy) = D(x)y + xD(y)$, a Lie subalgebra of $\mathrm{End}_k(A)$ under the commutator $[D, E] = D \circ E - E \circ D$. For $a \in A$, $\mathrm{ad} \ a : x \mapsto ax - xa$ is a derivation (`ad_mem_Der`). $\mathbf{Inn} \ k \ A = \{\mathrm{ad} \ a : a \in A\}$ is a Lie ideal of $\mathbf{Der} \ k \ A$ (the ideal property is $[D, \mathrm{ad} \ a] = \mathrm{ad}(Da)$). Finally $\mathbf{HH1} \ k \ A := \mathbf{Der}(A)/\mathbf{Inn}(A)$ with Mathlib’s quotient Lie algebra structure. Mathlib’s `Derivation` requires a commutative algebra, so these are defined directly. Solvability is Mathlib’s `LieAlgebra.IsSolvable`: some term of the derived series $L^{(0)} = L$, $L^{(m+1)} = [L^{(m)}, L^{(m)}]$ is 0.

This is the standard description of $\mathrm{HH}^1(A) = \mathrm{HH}^1(A, A)$: [LRW, Section 2] says (math symbols stripped by the retrieval tool) “the elements in () can be interpreted as the -derivations of into and the elements in () can be interpreted as the inner -derivations of into . It is well-known that () is a Lie algebra under the Lie bracket and () is a Lie ideal of (), so that () has a Lie algebra structure.”

Definition 2 (The quiver and the cycle condition). $V = \{1, 2\}$ with Mathlib `Quiver` structure `Hom 1 2 = Fin 3` and all other hom types empty. A path p visits v if v is its source or the target of one of its arrows. `CyclesBrokenByAtMostOneVertex W` means: there is a set S of vertices with at most one element (`S.Subsingleton`) such that every closed path of positive length visits some vertex of S ; equivalently, deleting S leaves no oriented cycle.

Definition 3 (The path algebra kK_3). P is the set of paths of K_3 : trivial paths e_1, e_2 and the arrows α_i . `paths_bijective` proves that `P.toPath` is a bijection onto all paths $\Sigma a b$, `Quiver.Path a b` of the Mathlib quiver. `PathAlg k = k^P` is the vector space with basis P , with the product given on basis elements by concatenation (`basis_mul_basis`): $e_1 e_1 = e_1$, $e_2 e_2 = e_2$, $e_1 \alpha_i = \alpha_i = \alpha_i e_2$, and every other product of two paths is 0 (the paths are not composable). Paths compose left to right, as in Mathlib’s `Quiver.Path.comp`. This is the path algebra as defined in [Wiki]: “multiplication given by concatenation of paths. If two paths cannot be concatenated because the end vertex of the first is not equal to the starting vertex of the second, their product is defined to be zero. This defines an associative algebra over”. The ring and k -algebra axioms are proved in Lean.

3 Proof

Lemma 4 (`K3_no_oriented_cycle`, `K3_cyclesBroken`). *Every closed path in K_3 has length 0. Hence K_3 satisfies the cycle condition with $S = \emptyset$.*

Proof. By induction on paths (`path_shape`), every path has length 0, or goes from 1 to 2 and has length 1; there is no arrow out of vertex 2. A closed path of length 1 would need $1 = 2$. $\square$

For $i \neq j$ in $\{0, 1, 2\}$ let D_{ij} (`Dlin i j`, `D i j`) be the linear map with $D_{ij}(\alpha_j) = \alpha_i$ and D_{ij} zero on e_1, e_2 and on the other arrows. In coordinates, $D_{ij}(x)$ has α_l -coordinate x_{α_j} if $l = i$ and 0 otherwise, and zero e -coordinates.

Lemma 5 (`Dlin_mem`). *Each D_{ij} is a derivation of kK_3 .*

Proof. The α_l -coordinate of xy is $x_{e_1} y_{\alpha_l} + x_{\alpha_l} y_{e_2}$ and the e -coordinates of xy involve only e -coordinates. Both sides of $D(xy) = D(x)y + xD(y)$ then have α_l -coordinate $[l = i](x_{e_1} y_{\alpha_j} + x_{\alpha_j} y_{e_2})$ and zero e -coordinates. $\square$

Lemma 6 (`D_bracket`). *For distinct i, j, m : $[D_{im}, D_{mj}] = D_{ij}$ in $\mathrm{Der}(kK_3)$.*

Proof. $D_{im} D_{mj}$ sends $\alpha_j \mapsto \alpha_i$ and kills everything else; $D_{mj} D_{im}$ is 0 because D_{im} has image in $k\alpha_i$ and $D_{mj}(\alpha_i) = 0$ for $i \neq j$. $\square$

Lemma 7 (`D_not_inner`). *For $i \neq j$, $D_{ij} \notin \mathrm{Inn}(kK_3)$.*

Proof. For any a , the α_i -coordinate of $(\mathrm{ad} a)(\alpha_j) = a\alpha_j - \alpha_j a$ is $a_{e_1} \cdot 0 + a_{\alpha_i} \cdot 0 - 0 \cdot a_{\alpha_i} - 0 \cdot a_{e_2} = 0$, while that of $D_{ij}(\alpha_j) = \alpha_i$ is 1. $\square$

Theorem 8 (`HH1_not_solvable`). *For every field k , the Lie algebra $\mathrm{HH}^1(kK_3) = \mathrm{Der}(kK_3) / \mathrm{Inn}(kK_3)$ is not solvable.*

Proof. Let d_{ij} ($d \ i \ j$) be the class of D_{ij} . The quotient map is a Lie homomorphism, so $[d_{im}, d_{mj}] = d_{ij}$ for distinct i, j, m (`d_bracket`). By induction on n , every d_{ij} with $i \neq j$ lies in the n -th derived term (`d_mem_derivedSeries`): for the step pick $m \notin \{i, j\}$, then $d_{ij} = [d_{im}, d_{mj}]$ with both factors in the n -th term. If the n -th term were 0, then $d_{01} = 0$, i.e. $D_{01} \in \text{Inn}$, contradicting `D_not_inner`. No characteristic assumption is used. $\square$

Theorem 9 (`conjecture_00000004044_false`). *CyclesBrokenByAtMostOneVertex* $V \wedge \neg \text{LieAlgebra.IsSolvable } (\text{HH1 } k \text{ (PathAlg } k))$ for every field k . Consequently (`conjecture_00000004044_iff_fails`) $\neg (\text{LieAlgebra.IsSolvable } (\text{HH1 } k \text{ (PathAlg } k))) \leftrightarrow \text{CyclesBrokenByAtMostOneVertex } V$.

Remark 10 (Scope and conventions). (1) What is refuted: the “solvable” reading, for quiver algebras meaning either the path algebra kQ or a bound quiver algebra kQ/I with I admissible (both include kK_3), over any fixed field. The “only if” direction is not addressed (one direction failing refutes the biconditional). (2) The Chinese body says *ke yue* (“reducible”) where the English body and the Chinese title say solvable. We read solvable as the intended statement: two of the three texts say so, and the conjecture’s own parenthetical name is “solvability criterion”. “Reducible Lie algebra” is not a standard notion; under the reading “reducible = has a nonzero proper ideal” the statement is *not* refuted here ($\text{HH}^1(kK_3) \cong \mathfrak{sl}_3$ is simple in characteristic 0, but proving that needs the full HH^1 computation and is not formalized). (3) Conventions: paths compose left to right; HH^1 uses the commutator bracket on $\text{End}_k(A)$ (Mathlib’s `LieRing.ofAssociativeRing`, enabled as a local instance as in Mathlib’s own Lie theory files); “removing at most one vertex” allows removing none.

4 Lean audit

File `lean/Conjecture4044/Basic.lean` (363 lines, only import Mathlib), Lean v4.33.1, Mathlib v4.33.1. Main theorem `C4044.conjecture_00000004044_false`. `#print` axioms for `conjecture_00000004044_false`, `conjecture_00000004044_iff_fails` and `paths_bijective` gives `[propext, Classical.choice, Quot.sound]`; no sorry, no native_decide, no added axioms.

References

- [LRW] Y. Liu, L. Rubio y Degraffi, C. Wen, *On the first Hochschild cohomology of finite dimensional quiver algebras under gluing idempotents*, <https://arxiv.org/abs/2211.05435v3>. On the Kronecker quiver (math stripped by retrieval): “if is the -Kronecker quiver (the quiver with parallel arrows), with the convention that -Kronecker quiver is the Dynkin quiver where is defined as . In other words, is the quotient by its center.”
- [Wiki] Quiver (mathematics), [https://en.wikipedia.org/wiki/Quiver_\(mathematics\)](https://en.wikipedia.org/wiki/Quiver_(mathematics)).